Technical Design Document (TDD)

Project: ThreadFlow (Production Order Tracking & Early Delay Warning SaaS)
Client: Company A, Tiruppur • **GitHub:** https://github.com/thanishmamilla/knitwear
Live Demo: https://knitwear-git-main-mamilla-thanish-kumars-projects.vercel.app/

---

1. Executive Summary & Financial Problem Formulation

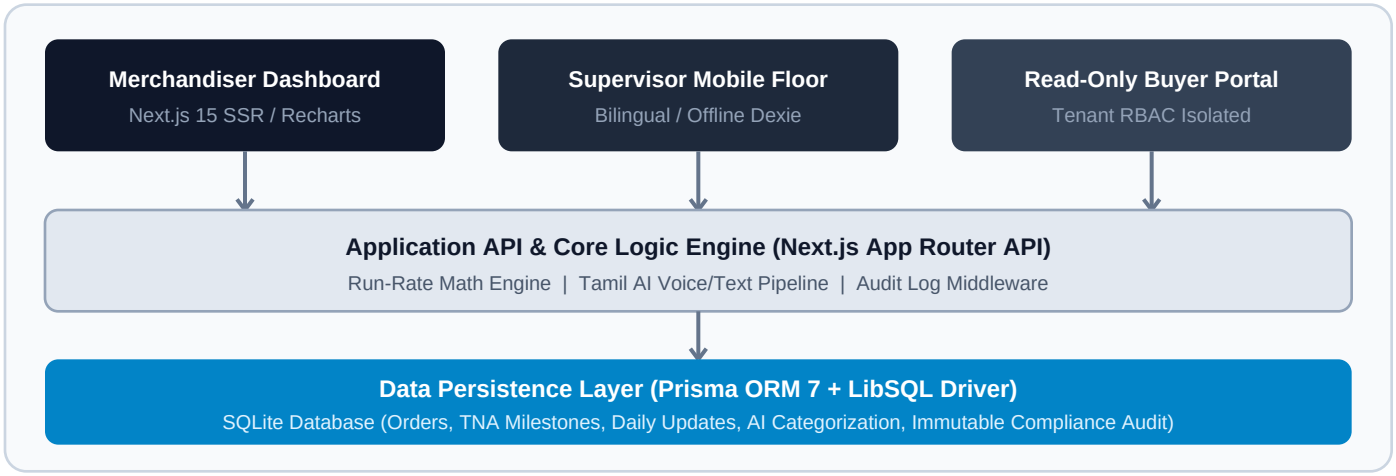
Company A is a ■95–110 crore Tiruppur-based knitwear exporter to UK and EU buyers with 3 owned manufacturing units plus subcontracted printing and washing (900 workers, 35 office staff). Currently, merchandisers track ~500 orders/year across multiple spreadsheets by phoning floor supervisors daily.

• **Financial Problem:** 8% (40 orders) ship late. 15 require air freight penalties (■52.5 lakh), 25 suffer 5% buyer discounts (■26.25 lakh), and 12 merchandisers waste 2 hours/day chasing status (7,200 hrs/yr × ■400 = ■28.8 lakh). **Total annual loss: ■1.08 Crore.**

• **The Solution:** A multi-tenant SaaS delivering floor-level daily updates in Tamil/English, transparent mathematical run-rate delay projections 2 weeks early, immutable buyer compliance change tracking, and isolated buyer portals.

2. System Architecture & Tech Stack

Tech Stack Choices: Next.js 15 (App Router, Server Actions, React 19) provides hybrid server-side rendering and client interactivity. SQLite with Prisma ORM 7 (`@prisma/adapter-libsql`) eliminates external database setup hurdles for local verification. Dexie.js (IndexedDB) supports offline updates for patchy factory network environments.



3. Relational Data Model (Schema Design)

Entity	Core Business Purpose	Key Attributes & Constraints
Order	Primary production order record	poNumber (unique), buyerId, unitId, style, color, totalQuantity, exFactoryDate, currentStage, currentProducedQty, riskStatus (GREEN/AMBER/RED)
Milestone	6-stage Time-and-Action (TNA) schedule	orderId, stage (Cutting → Printing → Stitching → Washing → Finishing → Packing), sequence, targetDate, completedDate, status
DailyUpdate	Floor quantity & observation entry	orderId, unitId, stage, quantityCompleted, photoUrl (base64 image), voiceUrl (audio), rawNote, language (ta/en), supervisorName
AIAnalysis	Dual-language governance record	dailyUpdateId, originalNote (Tamil), translation (English), predictedCategory, confidence, finalCategory, isOverridden, reviewedBy
DelayWarning	Run-rate alerts with buyer visibility gate	orderId, severity (AMBER/RED), actualRate, requiredRate, projectedDelayDays, isApprovedForBuyer (boolean gate), approvedBy
AuditLog	Immutable EU/UK compliance change log	orderId, field, oldValue, newValue, changedBy, reason (mandatory justification), timestamp

4. Mathematical Run-Rate Delay Engine (The Core Hard Part)

Instead of subjective merchandiser intuition, delays are flagged 2 weeks early via transparent throughput run-rate equations:
• **Remaining Quantity:** Total Order Quantity – Cumulative Produced Quantity
• **Days Left:** Ex-Factory Delivery Date – Today's Reference Date
• **Required Run-Rate:** Remaining Quantity ÷ Days Left (pieces per day required)
• **Actual Output Rate:** 7-day rolling average of completed stage output on the floor (pieces per day)
• **Projected Days Needed:** Remaining Quantity ÷ Actual Output Rate
• **Projected Delay Days:** Projected Days Needed – Days Left
Status Thresholds:
• **GREEN (On Schedule):** Actual Rate ≥ Required Rate (Delay ≤ 0 days)
• **AMBER (Early Warning):** 0 < Projected Delay ≤ 3.0 days (Early alert; internal floor intervention needed)
• **RED (Critical Risk):** Projected Delay > 3.0 days (Threatens air freight/discount; triggers merchandiser approval gate)

5. Tamil Voice Processing & AI Root Cause Pipeline

1. **Audio Floor Capture:** Standard HTML5 MediaRecorder captures voice notes on cheap Android devices (2–3 GB RAM).
2. **Bilingual Transcription:** Native speech recognition extracts Tamil (ta-IN) or English (en-US) into text with instant audio playback.
3. **Dual-Column Auditability:** Original Tamil text is preserved permanently beside the English translation to eliminate translation disputes.
4. **Root Cause Classification:** Groq Llama 3.3 70B (with deterministic shopfloor NLP fallback) classifies notes into 8 root cause categories: *Machine Breakdown, Material Delay, Quality Issue, Labour Shortage, Printing Delay, Washing Delay, Approval Pending, Other*.
5. **Merchandiser Approval Gate:** Merchandisers can override incorrect AI categories. Crucially, warnings reach the buyer portal **only** after explicit merchandiser confirmation.

6. Key Screens & Security Isolation

• **Order Board (`/`):** Minimalist table with status color badges, buyer/unit filters, and a 'Math Proof' modal detailing exact formula steps.
• **Order 360° Details (`/orders/[id]`):** Full view with 6-stage TNA timeline, size matrix, delay warning gate, and compliance history.
• **Supervisor Floor Entry (`/supervisor`):** Mobile-first ergonomics with large numeric buttons, 1-tap Tamil quick chips, voice recording player, and Dexie offline sync.
• **Multi-Tenant Buyer Portal (`/buyer`):** Strict isolation prevents Company B from seeing Company C orders, styles, or unapproved warnings.
• **Compliance Audit Trail (`/audit`):** Immutably records every ex-factory date adjustment with mandatory reasons for EU/UK compliance.

7. Scoping, Assumptions & Future Roadmap

• **Assumptions:** 6-day production work week in Tiruppur textile cluster; completed TNA milestones retain historical dates during adjustments.
• **Out of Scope (By Design):** Fabric inventory, piece-rate worker payroll, customs export documentation (per client brief).
• **Future Enhancements:** Automated WhatsApp Twilio bot for supervisors without mobile browsers; IoT stitch-counting sensors on sewing machines.